

Credit Card Approval Prediction

Introduction

Credit Card Approval Prediction is a machine learning project that predicts whether a customer's credit card application will be approved or rejected. The system uses customer information such as income and age to generate prediction results.

Project Description

The project uses a trained machine learning model to analyze customer details and predict the approval status. A Flask web application provides a simple interface for users to enter data and view the prediction.

Objectives

- Predict credit card approval using machine learning.
- Develop a simple Flask web application.
- Reduce manual effort in the approval process.
- Provide quick and accurate prediction results.

Technologies Used

- Python
- Flask
- Pandas
- Scikit-learn
- Joblib
- HTML

Dataset

The project uses a CSV dataset containing customer information for training the machine learning model. The dataset includes the following attributes:

- Income
- Age
- Approval Status

Installation

1. Install the required libraries using: `pip install -r requirements.txt`
2. Train the model by running: `python model.py`
3. Start the Flask application by running: `python app.py`
4. Open `http://127.0.0.1:5000` in your web browser.

Project Workflow

The dataset is loaded and used to train the machine learning model. The trained model is connected to the Flask application. Users enter customer details, and the system predicts whether the credit card application is approved or rejected.

Advantages

- Easy to use.
- Fast prediction results.
- Simple implementation.
- Suitable for academic projects.

Future Scope

The project can be enhanced by adding more customer features and advanced machine learning algorithms. It can also be integrated with real banking systems.

Conclusion

This project demonstrates the use of machine learning for credit card approval prediction. It provides quick and reliable results through a simple Flask web application.

Project Team

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